Electric Field

The space around the test charge in which it experiences a force is know as electric field.

We first place a positive charge q_0, called a test charge, at the point p. We then measure the electrostatic force that acts on the test charge. Finally, we define the electric field at point P due to the charged object as

$$\vec{E}=\vec{F}/q_0$$

Thus, the magnitude of the electric field $\vec{E}$ at point P is $E = F/q_0$, and the direction of is that of the force that acts on the positive test charge. SI unit is N/C.

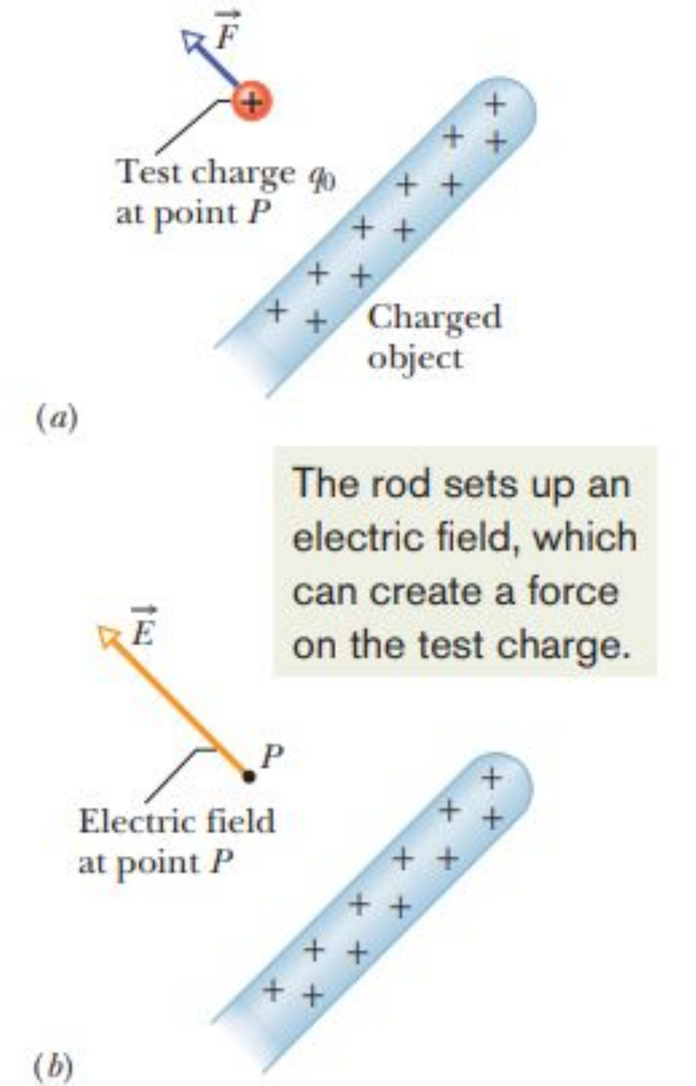


Fig. 22-1 (*a*) A positive test charge q_0 placed at point P near a charged object. An electrostatic force $\vec{F}$ acts on the test charge. (*b*) The electric field $\vec{E}$ at point P produced by the charged object.

Electric field due to a point charge

To find the electric field due to a charge q (or charged particle) at any point at a distance r from the charge, we put a positive test charge q_0 at that point. From Coulomb's law, the electrostatic force acting on q_0 is

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\,\hat{r}.$$

The direction of is directly away from the point charge if q is positive, and directly toward the point charge if q is negative. The electric field vector is

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\,\hat{r}$$

The direction of is the same as that of the force on the positive test charge: directly away from the point charge if q is positive, and toward it if q is negative

We can quickly find the net, or resultant, electric field due to more than one point charge. If we place a positive test charge q_0 near n point charges q_1, q_2,..., q_n, then the net force from the n point charges acting on the test charge is

$$\vec{F}_0 = \vec{F}_{01} + \vec{F}_{02} + \cdots + \vec{F}_{0n}.$$

the net electric field at the position of the test charge is

$$\begin{aligned}\vec{E} &= \frac{\vec{F}_0}{q_0} = \frac{\vec{F}_{01}}{q_0} + \frac{\vec{F}_{02}}{q_0} + \cdots + \frac{\vec{F}_{0n}}{q_0} \\ &= \vec{E}_1 + \vec{E}_2 + \cdots + \vec{E}_n.\end{aligned}$$

This shows us that the principle of superposition applies to electric fields as well as to electrostatic forces.

3. The nucleus of a plutonium-239 atom contains 94 protons. Assume that the nucleus is a sphere with radius 6.64 fm and with the charge of the protons uniformly spread through the sphere. At the nucleus surface, what are the (a) magnitude and (b) direction (radially inward or outward) of the electric field produced by the protons?

Electric field due to a dipole

$$E = \frac{1}{2\pi\varepsilon_0}\frac{p}{z^3}$$

Inspection of Eq shows that if you double the distance of a point from a dipole, the electric field at the point drops by a factor of 8. If you double the distance from a single point charge, however the electric field drops only by a factor of 4. Thus the electric field of a dipole decreases more rapidly with distance than does the electric field of a single charge. The physical reason for this rapid decrease in electric field for a dipole is that from distant points a dipole looks like two equal but opposite charges that almost—but not quite—coincide. Thus, their electric fields at distant points almost—but not quite—cancel each other.

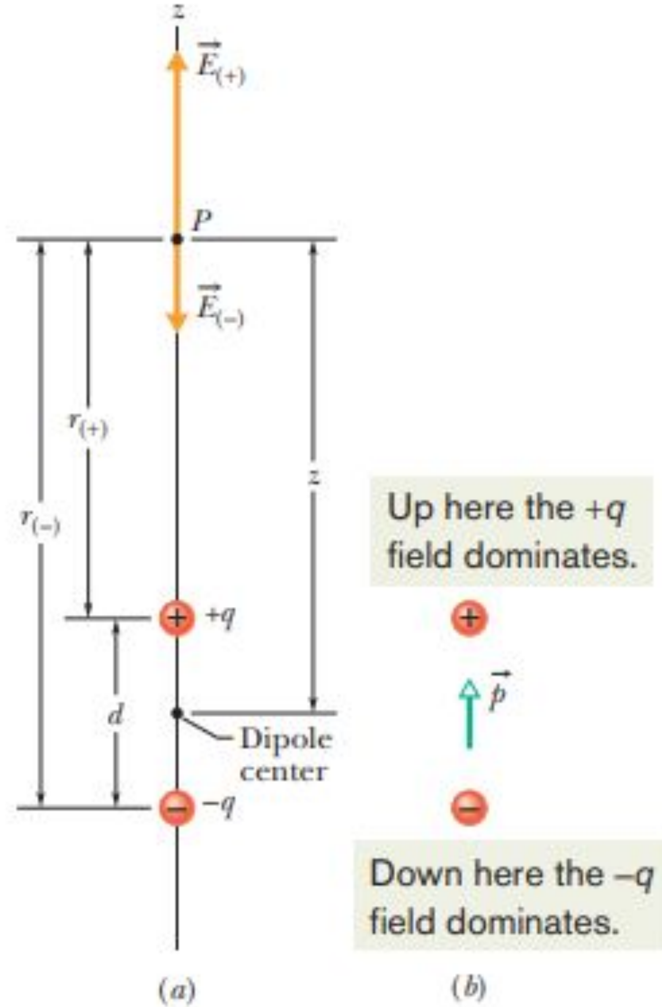
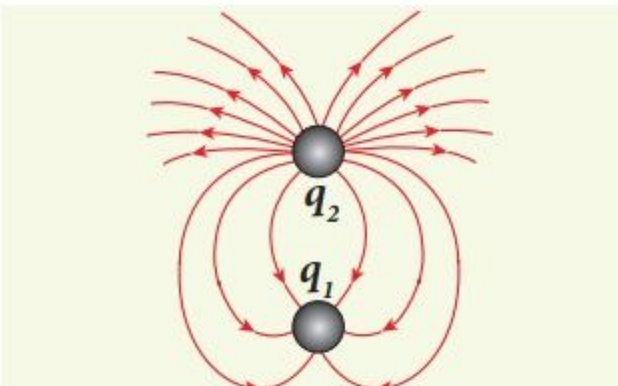
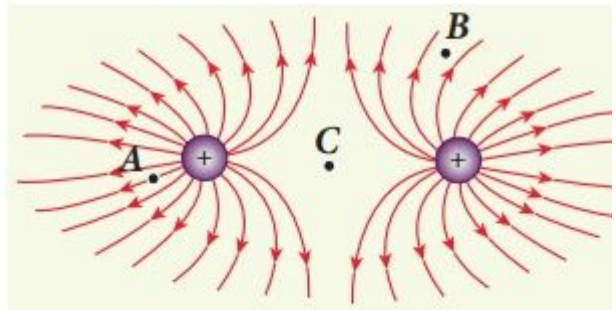


Fig. 22-8 (*a*) An electric dipole. The electric field vectors $\vec{E}_{(+)}$ and $\vec{E}_{(-)}$ at point P on the dipole axis result from the dipole's two charges. Point P is at distances $r_{(+)}$ and $r_{(-)}$ from the individual charges that make up the dipole. (*b*) The dipole moment $\vec{p}$ of the dipole points from the negative charge to the positive charge.

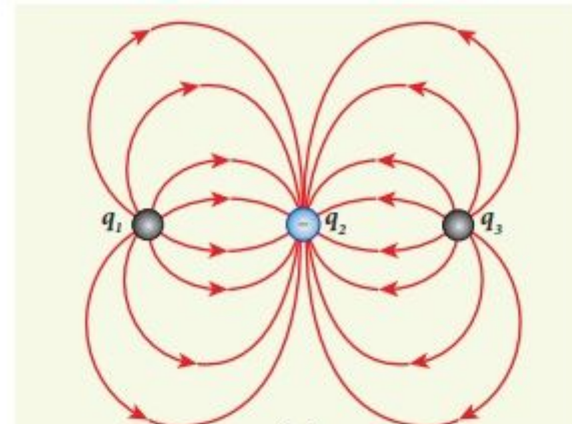
The following pictures depict electric field lines for various charge configurations.



(a)



(b)



(c)

(i) In figure (a) identify the signs of two charges and find the ratio |q1 / q2|

(ii) In figure (b), calculate the ratio of two positive charges and identify the strength of the electric field at three points A, B, and C

(iii) Figure (c) represents the electric field lines for three charges. If q2 = -20 nC, then calculate the values of q1 and q3

(i) The electric field lines start at q2 and end at q1. In figure (a), q2 is positive and q1 is negative. The number of lines starting from q2 is 18 and number of the lines ending at q1 is 6. So q2 has greater magnitude. The ratio of $|q1/ q2| = N1/ N2 = 6/18 = 1/3$. It implies that $|q2| = 3|q1|$

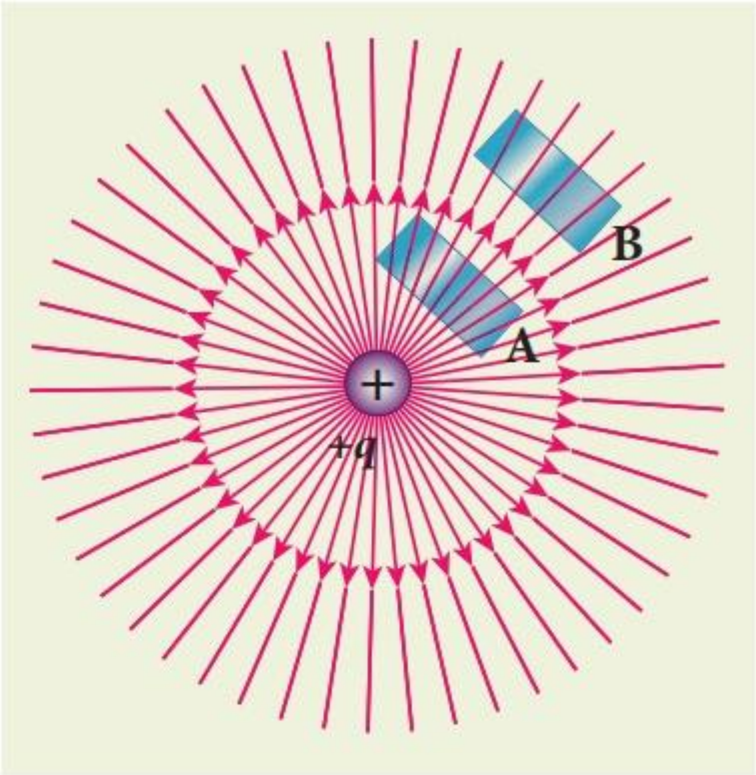
(ii) In figure (b), the number of field lines emanating from both positive charges are equal (N=18). So the charges are equal. At point A, the electric field lines are denser compared to the lines at point B. So the electric field at point A is greater in magnitude compared to the field at point B. Further, no electric field line passes through C, which implies that the resultant electric field at C due to these two charges is zero.

(iii) In the figure (c), the electric field lines start at q1 and q3 and end at q2. This implies that q1 and q3 are positive charges. The ratio of the number of field lines is $| q1 / q2 | = 8/16 = | q3 / q2 | = 1/2$, implying that q1and q3are half of the magnitude of q2. So q1 = q3 = +10 nC

Electric Flux

The number of electric field lines intersecting a given area is called electric flux. It is usually denoted by the Greek letter Φ_E and its unit is N m^2 C^{-1}.

The electric field of a point charge is drawn in this figure. Consider two small rectangular area elements placed normal to the field at regions A and B. Even though these elements have the same area, the number of electric field lines crossing the element in region A is more than that crossing the element in region B. Therefore the electric flux in region A is more than that in region B. The electric field strength for a point charge decreases as the distance increases, then for a point charge electric flux also decreases as the distance increases. The above discussion gives a qualitative idea of electric flux.



Electric flux for uniform Electric field

Consider a uniform electric field in a region of space. Let us choose an area A normal to the electric field lines as shown in Figure (a). The electric flux for this case is

$$\Phi_E = EA \qquad (1.52)$$

Suppose the same area A is kept parallel to the uniform electric field, then no electric field lines pierce through the area A , as shown in Figure (b). The electric flux for this case is zero.

$$\Phi_E = 0 \qquad (1.53)$$

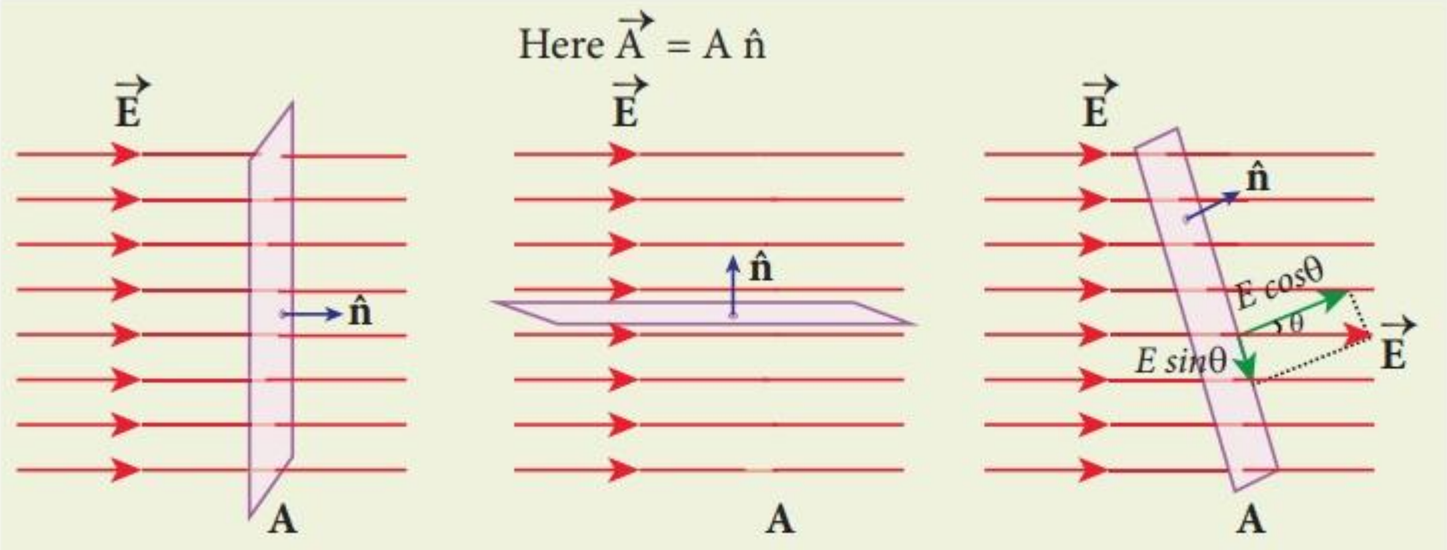
If the area is inclined at an angle θ with the field, then the component of the electric field perpendicular to the area alone contributes to the electric flux. The electric field component parallel to the surface area will not contribute to the electric flux. This is shown in Figure (c). For this case, the electric flux

$$\Phi_E = (E\cos\theta)\,A \qquad (1.54)$$

Further, θ is also the angle between the electric field and the direction normal to the area. Hence in general, for uniform electric field, the electric flux is defined as

$$\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta \qquad (1.55)$$

Here, note that $\vec{A}$ is the area vector $\vec{A}$ =A $\hat{n}$. Its magnitude is simply the area A and the direction is along the unit vector *n* perpendicular to the area as shown in Figure



(a) Electric flux = EA (b) Electric flux = 0 (c) Electric flux = $(E\cos\theta)A$

Flux through a closed cylinder, uniform field

Figure 23-4 shows a Gaussian surface in the form of a cylinder of radius R immersed in a uniform electric field , with the cylinder axis parallel to the field. What is the flux of the electric field through this closed surface?

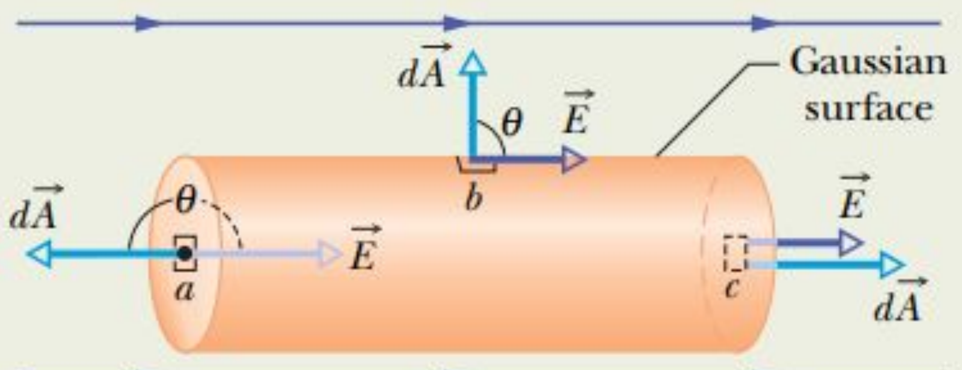


Fig. 23-4 A cylindrical Gaussian surface, closed by end caps, is immersed in a uniform electric field. The cylinder axis is parallel to the field direction.

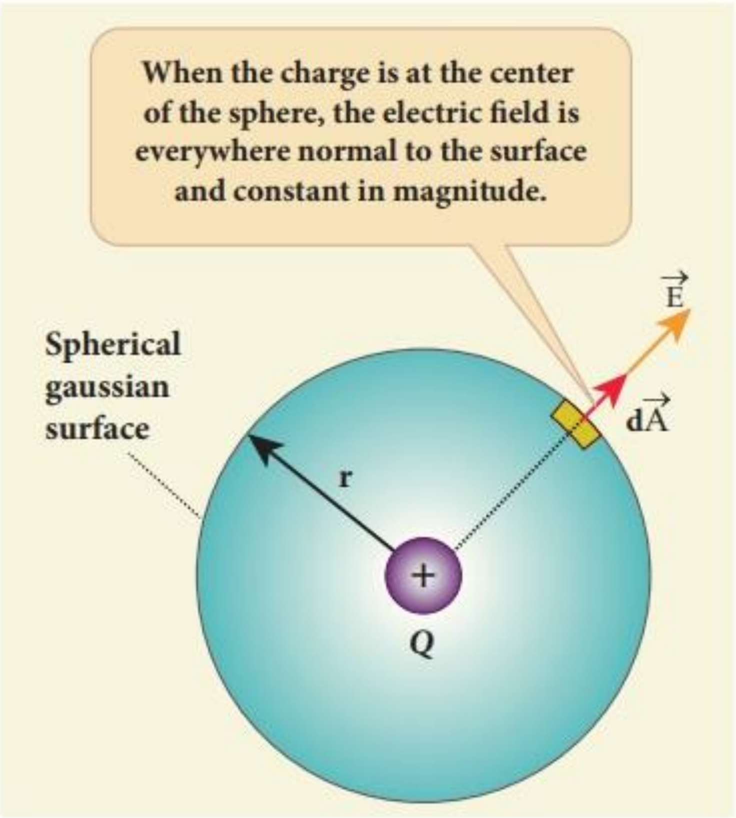
Gauss law

Gauss's law states that **if a charge Q is enclosed by an arbitrary closed surface, then the total electric flux** Φ_E **through the closed surface is**

$$\Phi_E = {q_{enc}}/{\epsilon_0}$$

The equation is called as Gauss's law.

To arrive at equation, we have chosen a spherical surface. This imaginary surface is called a Gaussian surface. The shape of the Gaussian surface to be chosen depends on the type of charge configuration and the kind of symmetry existing in that charge configuration. The electric field is spherically symmetric for a point charge, therefore spherical Gaussian surface is chosen. Cylindrical and planar Gaussian surfaces can be chosen for other kinds of charge configurations.



5. In Fig. a proton is a distance d/2 directly above the center of a square of side d. What is the magnitude of the electric flux through the square? (Hint: Think of the square as one face of a cube with edge d.)

7. A point charge of 1.8 μC is at the center of a Gaussian cube 55 cm on edge. What is the net electric flux through the surface?

Application of Gauss's theorem

Electric field intensity due to thin infinite long wire (Cylindrical symmetry)

Figure 23-12 shows a section of an infinitely long cylindrical plastic rod with a uniform positive linear charge density l. Let us find an expression for the magnitude of the electric field at a distance r from the axis of the r

We choose a circular cylinder of radius r and length h, coaxial with the rod. Because the Gaussian surface must be closed, we include two end caps as part of the surface.

At every point on the cylindrical part of the Gaussian surface, $\vec{E}$ must have the same magnitude E and (for a positively charged rod) must be directed radially outward.

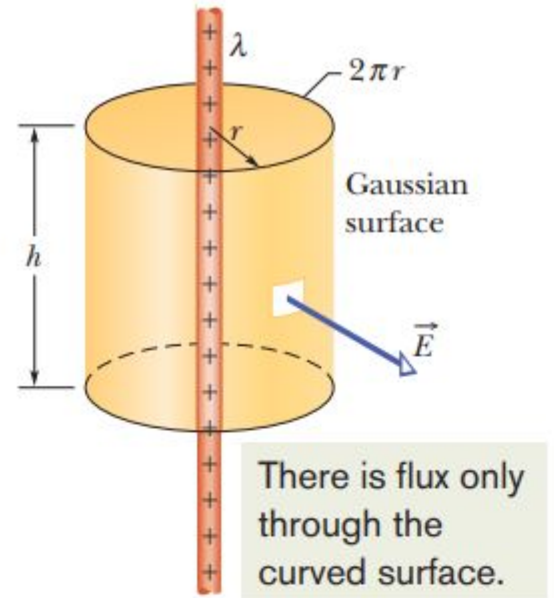


Fig. 23-12 A Gaussian surface in the form of a closed cylinder surrounds a section of a very long, uniformly charged, cylindrical plastic rod.

Since $2\pi r$ is the cylinder's circumference and h is its height, the area A of the cylindrical surface is $2\pi rh$. The flux through this cylindrical surface is then

$$\Phi = EA\cos\theta = E(2\pi rh)\cos 0 = E(2\pi rh)$$

There is no flux through the end caps because , E being radially directed, is parallel to the end caps at every point.

The charge enclosed by the surface is λh, which means Gauss' law

$$\varepsilon_0\Phi = q_{enc},$$

reduces to

$$\varepsilon_0 E(2\pi rh) = \lambda h,$$

yielding

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} \quad \text{(line of charge).}$$

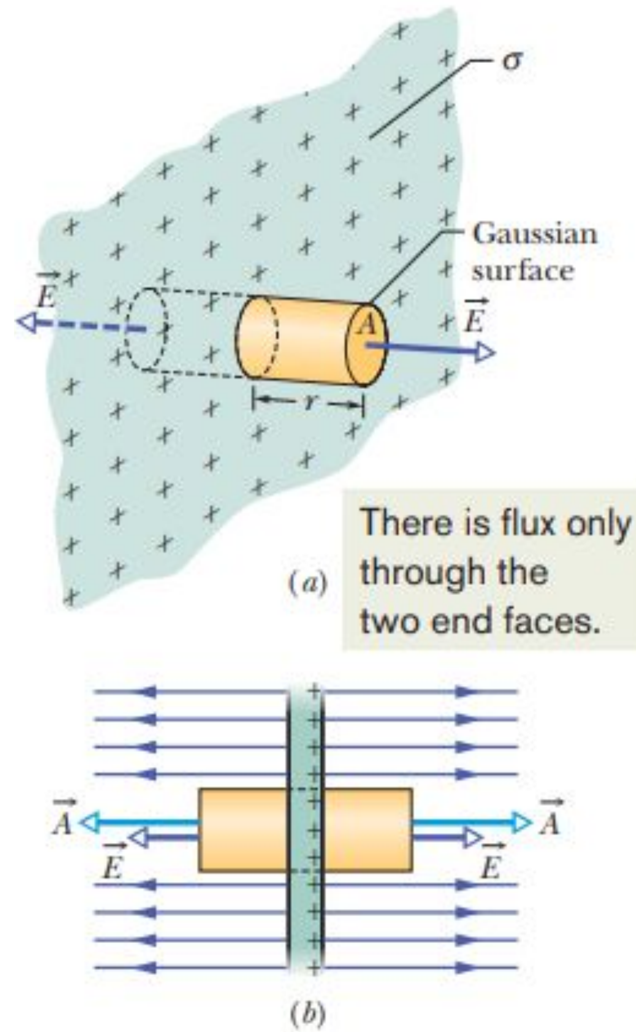
Application of Gauss's theorem

Electric field intensity due to thin infinite plane of sheet (Planar symmetry)

Figure shows a portion of a thin, infinite, nonconducting sheet with a uniform (positive) surface charge density σ. Let us find the electric field a distance r in front of the sheet.

A useful Gaussian surface is a closed cylinder with end caps of area A, arranged to pierce the sheet perpendicularly as shown. Since the charge is positive, $\vec{E}$ is directed away from the sheet, and thus the electric field lines pierce the two Gaussian end caps in an outward direction.

Because the field lines do not pierce the curved surface, there is no flux through this portion of the Gaussian surface



(a) Perspective view and (b) side view of a portion of a very large, thin plastic sheet

Gaussian surface. Thus $\vec{E} \cdot d\vec{A}$ is simply $E\,dA$; then Gauss' law,

$$\varepsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{\text{enc}},$$

becomes

$$\varepsilon_0(EA + EA) = \sigma A,$$

where σA is the charge enclosed by the Gaussian surface. This gives

$$E = \frac{\sigma}{2\varepsilon_0} \qquad \text{(sheet of charge).}$$

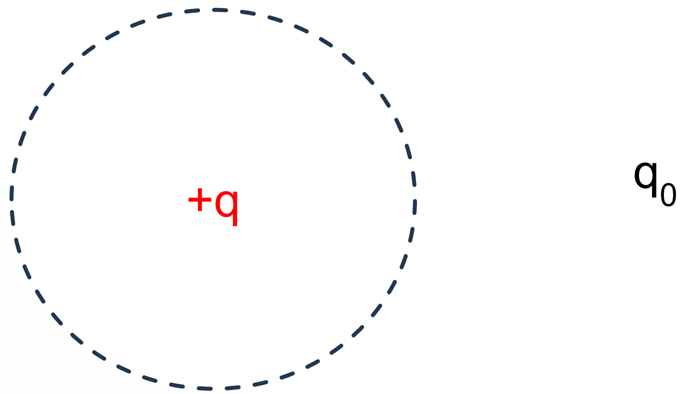
Since we are considering an infinite sheet with uniform charge density, this result holds for any point at a finite distance from the sheet

22. An electron is released 8.0 cm from a very long nonconducting rod with a uniform charge density 6.0 μ C/m. What is the magnitude of the electron's initial acceleration?

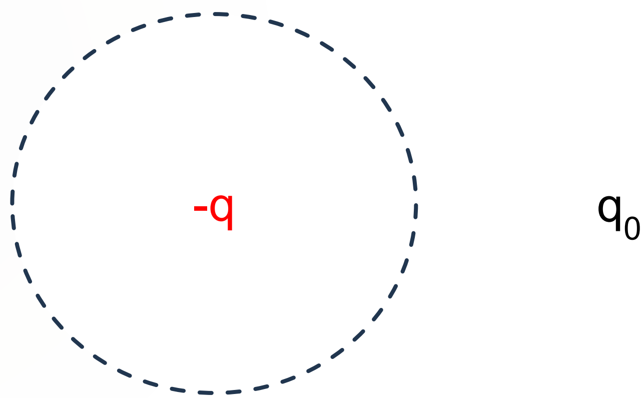
23 (a) The drum of a photocopying machine has a length of 42 cm and a diameter of 12 cm. The electric field just above the drum's surface is 2.3 105 N/C. What is the total charge on the drum? (b) The manufacturer wishes to produce a desktop version of the machine. This requires reducing the drum length to 28 cm and the diameter to 8.0 cm. The electric field at the drum surface must not change. What must be the charge on this new drum?

Electric potential

It is defined as the amount of work done to bring a unit charge from infinity to a certain point.



To bring a positive test charge within the field of +q charge work is required



To bring a positive test charge within the field of -q charge work is also required to keep it stable.

Potential energy: PE of a charge at a point is the work done by external force in bringing the charge from infinity to that point.

For an example of this, suppose we place a test particle of positive charge 1.60 $\times 10^{-19}$ C at a point in an electric field where the particle has an electric potential energy of 2.40 $\times 10^{17}$ J. Then the potential energy per unit charge is

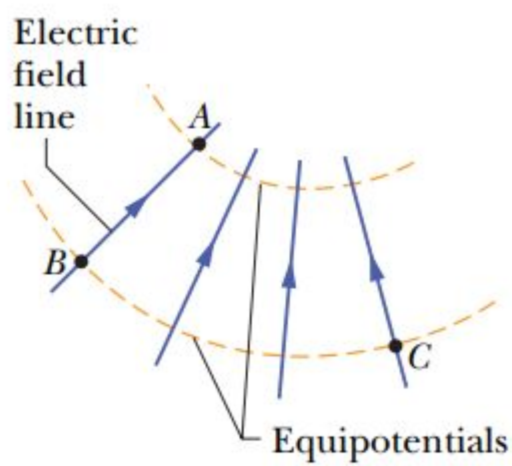
$$\frac{2.40\times10^{17}}{1.60\times10^{-19}} = 150 \text{ J/C}$$

Next, suppose we replace that test particle with one having twice as much positive charge, 3.20 $\times 10^{-19}$ C. We would find that the second particle has an electric potential energy of 4.80 $\times 10^{17}$ J, twice that of the first particle. However, the potential energy per unit charge would be the same, still 150 J/C.

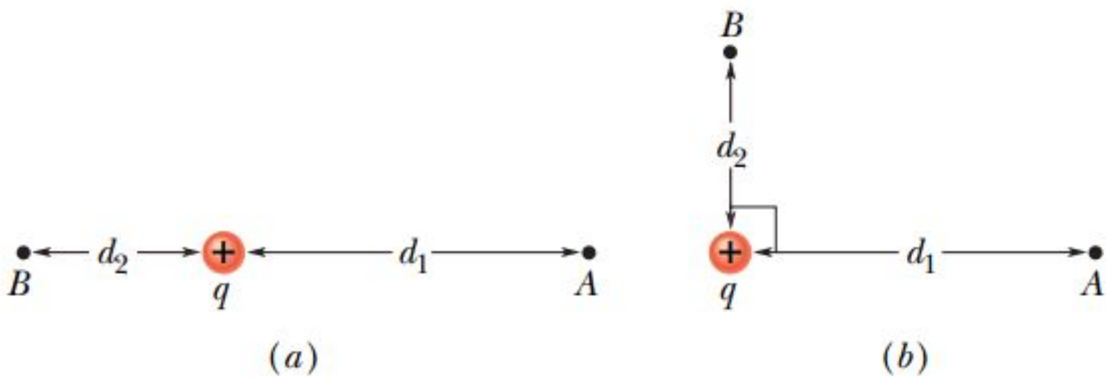
Thus, the potential energy per unit charge, which can be symbolized as U/q, is independent of the charge q of the particle and is characteristic only of the electric field we are investigating. The potential energy per unit charge at a point in an electric field is called **the electric potential V** (or simply the potential) at that point. V=U/q

Potential energy difference: PE difference between two points is the work required to be done by external force in moving a charge from one point to another.

6.When an electron moves from A to B along an electric field line in Fig., the electric field does 3.94×10^{-19} J of work on it. What are the electric potential differences (a) $V_B - V_A$, (b) $V_C - V_A$, and (c) $V_C - V_B$?

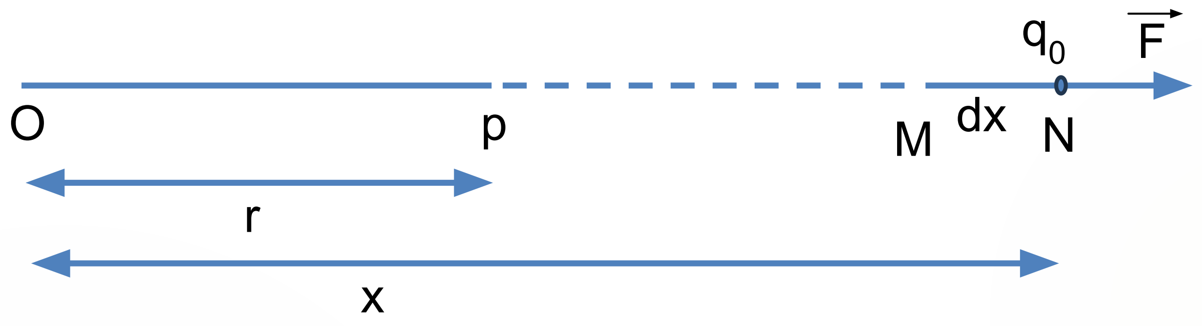


14.Consider a point charge q $1.0\ \mu C$, point A at distance d_1 2.0 m from q, and point B at distance d_2 1.0 m. (a) If A and B are diametrically opposite each other, as in Fig.a, what is the electric potential difference $V_A - V_B$? (b) What is that electric potential difference if A and B are located as in Fig.b?

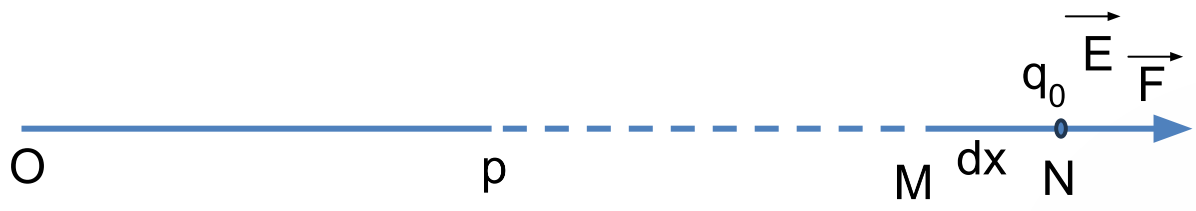


15.A spherical drop of water carrying a charge of 30 pC has a potential of 500 V at its surface (a) What is the radius of the drop? (b) If two such drops of the same charge and radius combine to form a single spherical drop, what is the potential at the surface of the new drop?

Potential due to a point charge



Relation between electric field and potential



The change in potential with distance is called potential gradient. Hence the electric field is equal to negative gradient of potential.

$$E = -\frac{dV}{dx}$$

The negative sign indicates that, the potential decreases in the Direction of electric field.